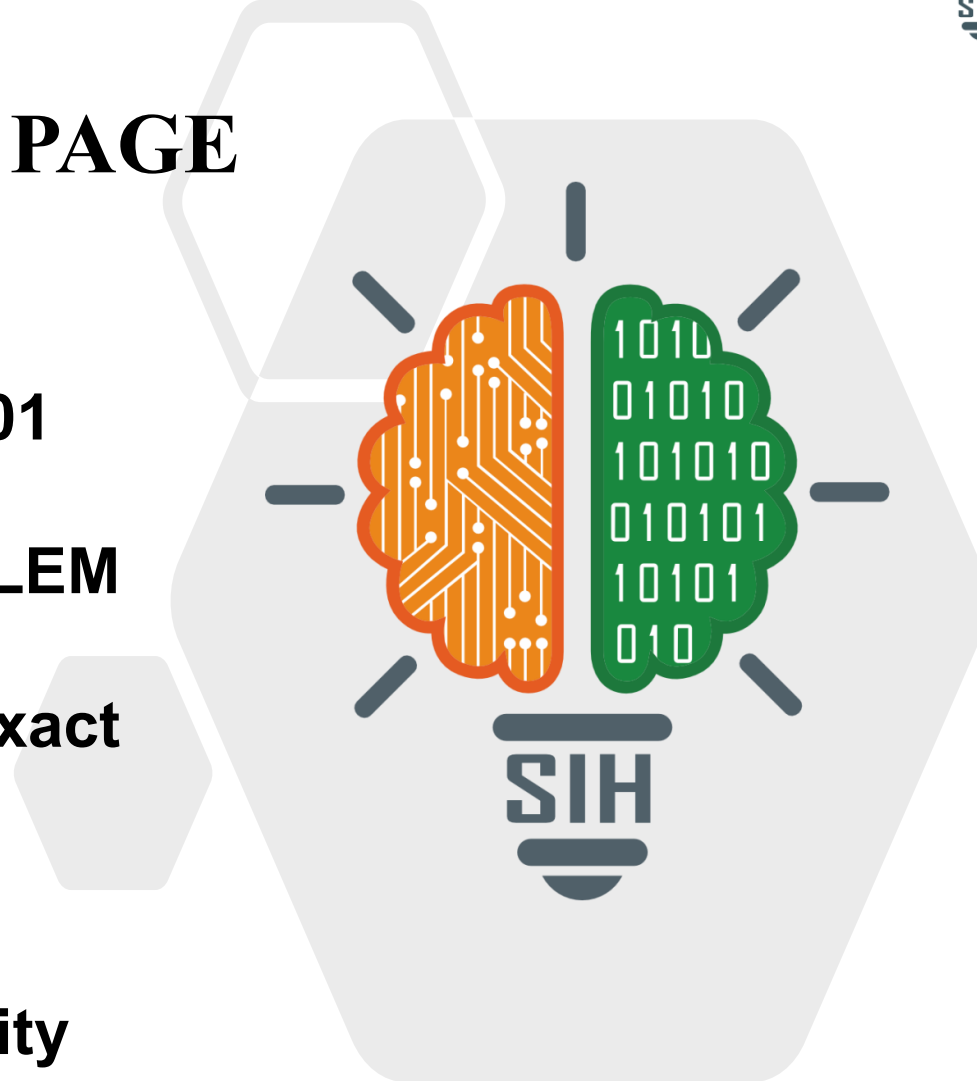


TITLE PAGE

- Problem Statement ID – STH260001
- Problem Statement Title- [PROBLEM STATEMENT TITLE — paste exact SIH portal wording]
- Theme- Blockchain & Cybersecurity
- PS Category- Hardware



❖ Proposed Solution

- **What it is:** A self-contained, offline-first edge node for comms, storage, sync and local AI — zero cloud dependency.
- **Mesh:** Local peer discovery (mDNS) + Ed25519/QUIC auth + an append-only signed event log — converges correctly across partitions, no blockchain overhead.
- **Trust:** Every peer is explicitly enrolled (UNKNOWN → PENDING → TRUSTED → REVOKED); enforced in the Rust core, never the frontend.
- **Local AI:** On-device Qwen2.5-1.5B + BGE embeddings via llama.cpp — zero API keys, zero network calls, citation-verified answers.
- **Proof, not pitch:** 499 automated tests, a working Tauri desktop app, and measured — not invented — evaluation numbers.

TECHNICAL APPROACH

- **Core:** Rust (Tauri 2) backend; React + TypeScript frontend; SQLite with versioned migrations.
- **Identity:** Ed25519 keypairs; node ID = SHA-256(public key); private key never leaves the core, never logged.
- **Networking:** libp2p over QUIC; mDNS discovery; TLS handshake reuses the node's own signing key — no custom crypto.
- **Data model:** Append-only signed event log, per-origin monotonic sequence numbers; conflicts flagged, never silently overwritten.
- **Local AI / RAG:** llama.cpp (CPU-only) running Qwen2.5-1.5B-Instruct + BGE-small-en-v1.5; schema-constrained JSON output; vector search over SQLite.
- **Methodology:** Test-first — 499 automated tests, clippy-clean, deterministic synthetic datasets, a reproducible AI benchmark harness.

FEASIBILITY AND VIABILITY



- **Feasible today:** Core mesh + local AI are already implemented and measured across 5 completed phases — not a concept.
- **Commodity hardware:** Runs on an ordinary laptop CPU (~2.5s per incident analysis, no GPU required) — realistic for field-deployable nodes.
- **Risk — model accuracy:** Small on-device models can misclassify; the model's output is advisory and never overwrites an operator's own severity rating.
- **Risk — revocation:** A trust decision is local today; Phase 2.75 adds signed revocation propagation across the mesh.
- **Risk — key storage:** The private key is currently a protected file, not hardware-backed; a hardware feasibility study (Jetson Orin + TrustZone/OP-TEE) already maps the migration path.
- **Strategy:** Every claim in this deck is backed by a passing test or a measured number — the same discipline scales toward deployment.

IMPACT AND BENEFITS

- **Who benefits:** Disaster-response teams, defence/field units, and remote operations in low- or no-connectivity environments.
- **Continuity when networks fail:** Incident capture, sync and AI-assisted triage keep working exactly when centralised systems go dark.
- **Security by default:** No cloud dependency means no exfiltration surface, no vendor key, no dependence on infrastructure the team doesn't control.
- **Faster, trustworthy decisions:** Local AI turns raw field reports into classified, cited incidents in ~2.5 seconds — no data centre, no wait.
- **Auditable, not blind, trust:** Every peer, event and trust decision is signed and independently verifiable — no black-box central authority.
- **Economic:** Runs on commodity edge hardware; no recurring cloud or API cost.

RESEARCH AND REFERENCES



- **Project repository:** github.com/Rengoku9000/Secure-Mesh-V1 — source, docs, tests and measured benchmarks.
- **Qwen2.5-1.5B-Instruct:** huggingface.co/Qwen/Qwen2.5-1.5B-Instruct-GGUF (Apache-2.0).
- **BGE-small-en-v1.5:** huggingface.co/CompendiumLabs/bge-small-en-v1.5-gguf (MIT).
- **llama.cpp:** github.com/ggml-org/llama.cpp (MIT), build b10375, CPU inference runtime.
- **libp2p:** libp2p.io — peer-to-peer networking stack (QUIC transport, mDNS discovery).
- **NVIDIA Jetson Linux Developer Guide:** docs.nvidia.com/jetson — OP-TEE, Secure Boot, Firmware TPM.
- **OP-TEE:** optee.readthedocs.io — open-source TrustZone trusted execution environment.